

Day 35 - Extended ACLs

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10:46 AM

ACL Info

ACL Configuration Methods

There are two ways of configuring ACLs. One is done through the 'global configuration' mode and the other is through the 'ACL configuration' mode where you write the configuration in sub-commands.

- **Global configuration** is done through the command:
 - `access-list NUMBER { permit | deny } IP WILDCARD-MASK`
- **ACL configuration** is done through the command:
 - `ip access-list standard NAME/NUMBER`

Advantages of Named ACL Config Mode

- Going into the **ACL configuration** mode allows you to delete individual entries in the **ACL** with `no ENTRY-NUMBER`. (Deleting individual entries in **global configuration** will lead to the deletion of the whole **ACL**, even if one entry was specified)
- You can insert new entries in between other entries by specifying the `sequence number`.

Resequencing ACLs

ACL has a **resequencing** function that allows you to edit ACL entries.

- It is done through the command:
 - `ip access-list resequence ACL-ID STARTING-SEQ-NUM INCREMENT`

Extended ACLs

Extended ACLs are very similar to Standard ACLs

- **Extended ACLs** match based on `source/destination IP address, source/destination port address`, etc.
- **Extended ACLs** can use the number range (`100 - 199`) and (`2000 - 2699`)
- They should be applied as close to the source as possible.
- They include two subtypes:
 1. **Extended numbered** ACLs
 2. **Extended named** ACLs

Extended ACL Configuration

There are two ways to create an extended ACL:

1. **Global configuration / directly**
 - This method only allows you to create **numbered ACLs**
 - Done through the command:
 - `access-list EXTENDED-ACL-NUMBER { permit | deny } PROTOCOL SRC-IP DST-IP`
2. **ACL configuration mode**
 - Done through the command:
 - `ip access-list extended NUMBER/NAME`
 - Which puts you in **Extended ACL configuration** mode. **ACEs** can be set through the following command format:
 - `[entry-number] { deny | permit } protocol src-ip dst-ip`
 - `[entry-number] { deny | permit } protocol src-ip argument source-port-num dst-ip argument dst-port-num`

Standard and Extended ACL - Host addresses

In standard ACLs, you could specify the `source IP address`, insert the command, and the router would automatically assume that it's a `host` route.

However, in extended ACLs, to specify a `/32` source or destination you have to use either the `host` option or specify the `wildcard mask`. You can't just write the address without either of those.

Extended ACL Protocols

Extended ACL Protocols

In extended ACLs you can filter by what protocol is being used in the packet. Some of the important protocols you need to know for the CCNA are these ones:

IP Protocol Numbers

Protocol	Number
ICMP	1
TCP	6
UDP	17
EIGRP	88
OSPF	89

[Link to original](#)

Extended ACL Protocol Arguments

- When matching **TCP/UDP**, you can optionally specify the *source* AND/OR *destination* ports to match.
- This is accomplished through the *protocol arguments* that can be written after '*src-ip*' or '*dst-ip*' while configuring **ACLs**

Argument	Effect	Example
eq (Equal)	Matches ports <i>equal</i> to the specified port	eq 80 equal to port 80
gt (Greater than)	Matches ports <i>higher</i> than the specified port	gt 80 greater than 80 (81 and greater)
lt (Less than)	Matches ports <i>lower</i> than the specified port	lt 80 less than 80 (79 and less)
neq (Not equal)	Matches ports that <i>aren't</i> the specified port	neq 80 not equal to 80
range	Matches ports that <i>fit</i> into the specified <i>range</i>	range 80 100 from port 80 to port 100

General note on Extended ACLs

If you specify the *protocol*, *source IP*, *source port*, *destination IP*, *destination port*, etc, a packet must match ALL of those values to match the ACL entry. Even if it matches all except one of the parameters, the packet won't match that entry of the ACL.